

Lab 1: Implementing RBAC

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

- Catalogs
- Event Triggers
- Extensions
- Foreign Data Wrappers
- Languages
- Publications
- Schemas (1)
- Subscriptions
- postgres
 - Login/Group Roles (21)
 - jon
 - mike
 - pg_checkpoint
 - pg_create_subscription
 - pg_database_owner
 - pg_execute_server_program
 - pg_maintain
 - pg_monitor
 - pg_read_all_data
 - pg_read_all_settings**
 - pg_read_all_stats
 - pg_read_server_files
 - pg_signal_autovacuum_process
 - pg_signal_backend

Query

```
1 CREATE ROLE cashier_role;  
2  
3 CREATE USER jack WITH PASSWORD 'jack123';  
4 CREATE USER bob WITH PASSWORD 'bob123';  
5  
6 GRANT cashier_role to jack;  
7 GRANT cashier_role to bob;  
8  
9 GRANT SELECT  
10
```

Data Output Messages Notifications

CREATE ROLE

Query returned successfully in 104 msec.

Total rows: Query complete 00:00:00.104

Query returned successfully in 104 msec. X

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Query

```
1 CREATE ROLE cashier_role;  
2  
3 CREATE USER jack WITH PASSWORD 'jack123';  
4 CREATE USER bob WITH PASSWORD 'bob123';  
5  
6 GRANT cashier_role to jack;  
7 GRANT cashier_role to bob;  
8  
9 GRANT SELECT  
10
```

Data Output Messages Notifications

CREATE ROLE

Query returned successfully in 109 msec.

Total rows: Query complete 00:00:00.109

Query returned successfully in 109 msec. X

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File Object Tools Edit View Window Help

Object Explorer: /postgres... x dvdrental/postgres@PostgreSQL 18* x public.customer/d... x public.rental/dvdre... x public.inventory/d...

Query: Query History

```
1 CREATE ROLE cashier_role;
2
3 CREATE USER jack WITH PASSWORD 'jack123';
4 CREATE USER bob WITH PASSWORD 'bob123';
5
6 GRANT cashier_role to jack;
7 GRANT cashier_role to bob;
8
9 GRANT SELECT
10
```

Data Output Messages Notifications

GRANT ROLE

Query returned successfully in 98 msec.

Total rows: Query complete 00:00:00.098

✓ Query returned successfully in 98 msec. ✕

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File Object Tools Edit View Window Help

Object Explorer: /postgres... x dvdrental/postgres@PostgreSQL 18* x public.customer/d... x public.rental/dvdre... x public.inventory/d...

Query: Query History

```
1 CREATE ROLE cashier_role;
2
3 CREATE USER jack WITH PASSWORD 'jack123';
4 CREATE USER bob WITH PASSWORD 'bob123';
5
6 GRANT cashier_role to jack;
7 GRANT cashier_role to bob;
8
9 GRANT SELECT ON film, staff TO cashier_role;
```

Data Output Messages Notifications

GRANT

Query returned successfully in 116 msec.

Total rows: Query complete 00:00:00.116

✓ Query returned successfully in 116 msec. ✕

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File Object Tools Edit View Window Help

Object Explorer: /postgres... x dvdrental/postgres@PostgreSQL 18* x public.customer/d... x public.rental/dvdre... x public.inventory/d...

Query: Query History

```
1 CREATE ROLE cashier_role;
2
3 CREATE USER jack WITH PASSWORD 'jack123';
4 CREATE USER bob WITH PASSWORD 'bob123';
5
6 GRANT cashier_role to jack;
7 GRANT cashier_role to bob;
8
9 GRANT SELECT ON film, staff TO cashier_role;
10 GRANT SELECT, INSERT ON payment TO cashier_role;
11
```

Data Output Messages Notifications

GRANT

Query returned successfully in 104 msec.

✓ Query returned successfully in 104 msec. ✕

Lab 2: Implementing RLS

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File Object Tools Edit View Window Help

Object Explorer: /postgres... x dvdrental/postgres@PostgreSQL 18* x public.customer/d... x public.rental/dvdre... x public.inventory/d...

Query: Query History

```
1 ALTER TABLE payment ENABLE ROW LEVEL SECURITY;
```

Data Output Messages Notifications

ALTER TABLE

Query returned successfully in 92 msec.

✓ Query returned successfully in 92 msec. ✕

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File Object Tools Edit View Window Help

Object Explorer: Catalogs, Event Triggers, Extensions, Foreign Data Wrappers, Languages, Publications, Schemas (1), Subscriptions, postgres, Login/Group Roles (21): jon, mike, pg_checkpoint, pg_create_subscription, pg_database_owner, pg_execute_server_program, pg_maintain, pg_monitor, pg_read_all_data, pg_read_all_settings, pg_read_all_stats, pg_read_server_files

Query: ALTER TABLE payment ENABLE ROW LEVEL SECURITY;
CREATE POLICY isolate_cashier_payments ON payment FOR SELECT TO cashier_role USING (staff_id = (SELECT staff_id FROM staff WHERE username = current_user));

Data Output: CREATE POLICY
Query returned successfully in 110 msec.

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File Object Tools Edit View Window Help

Object Explorer: Catalogs, Event Triggers, Extensions, Foreign Data Wrappers, Languages, Publications, Schemas (1), Subscriptions, postgres, Login/Group Roles (21): jon, mike, pg_checkpoint, pg_create_subscription, pg_database_owner, pg_execute_server_program, pg_maintain, pg_monitor, pg_read_all_data, pg_read_all_settings, pg_read_all_stats, pg_read_server_files

Query: ALTER TABLE payment ENABLE ROW LEVEL SECURITY;
CREATE POLICY isolate_cashier_payments ON payment FOR SELECT TO cashier_role USING (staff_id = (SELECT staff_id FROM staff WHERE username = current_user));
SET ROLE jack;
SELECT COUNT(*) FROM payment;

Data Output: count bigint
1 0

Showing rows: 1 to 1 of 1

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Query Editor

Query History

```
1 ALTER TABLE payment ENABLE ROW LEVEL SECURITY;
2
3 CREATE POLICY isolate_cashier_payments ON payment
4 FOR SELECT
5 TO cashier_role
6 USING (
7     staff_id = (SELECT staff_id FROM staff WHERE username = current_user)
8 );
9
10 SET ROLE jack;
11 SELECT COUNT(*) FROM payment;
12
13 SET ROLE bob;
14 SELECT COUNT(*) FROM payment;
```

Data Output

Showing rows: 1 to 1

Page No: 1 of 1

	count
1	0

Successfully run. Total query runtime: 147 msec. 1 rows affected.